RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:20 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:21 AM
Time taken	43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Divide and Conquer function to find the first occurrence index of 0
int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        // Find mid point to avoid integer overflow
        int mid = low + (high - low) / 2;

        // Check if mid element is the first 0
        // It is the first 0 if mid is 0 OR the element before it is 1
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }

        // If the element at mid is 1, then the first 0 must be in the right half
        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high);
        }

        // If the element at mid is 0 but it's not the first 0, look in the left half
        return findFirstZero(arr, low, mid - 1);
    }

    // If 0 is not found at all in the array
    return -1;
}

int main() {
    int m;

    // Read the size of the array
    if (scanf("%d", &m) != 1) {
        return 0;
    }

    // Dynamically allocate memory for the array
    int *arr = (int *)malloc(m * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read array elements
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    // Find the index of the first zero using Divide and Conquer
    int firstZeroIndex = findFirstZero(arr, 0, m - 1);

    // Calculate number of zeroes
    int zeroCount = 0;
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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